Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
8	(a)	(i)		(metals in Group 1) get more reactive (down the group) (1) award (1) for any of following due to a decrease in attraction between the nucleus and the outer shell electron easier to remove outer electron because there are more shells easier to remove outer electron because it is further from the nucleus	2			2		1
		(ii)		Group 1 metals are more reactive than Group 2 metals (1) award (1) for either of following • because Group 1 metals only need to lose 1 electron (from the outer shell) whereas Group 2 metals need to lose 2 electrons • because it is easier to lose 1 electron than 2 electrons	2			2		
	(b)			257.6 / 258 (3) if answer incorrect credit each correct step in one of two possible methods (ecf possible throughout) method 1 $n(H_2) = \frac{11.2}{2} = 5.6$ (1) $n(Na) = 5.6 \times 2 = 11.2$ (1) mass $Na = 11.2 \times 23 = 257.6$ (1) method 2 1 mol H_2 produced by 2 mol Na / 2 g H_2 produced by 46 g Na (1) 1 g H_2 produced by 23 g Na (1) 11.2 g H_2 produced by $23 \times 11.2 = 257.6$ g Na (1)		3		3	3	

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(c)			$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$ award (2) for correct equation award (1) if $Ca(OH)_2$ formula is correct		2		2		1
				Question 8 total	**4**	**5**	**0**	**9**	**3**	**2**